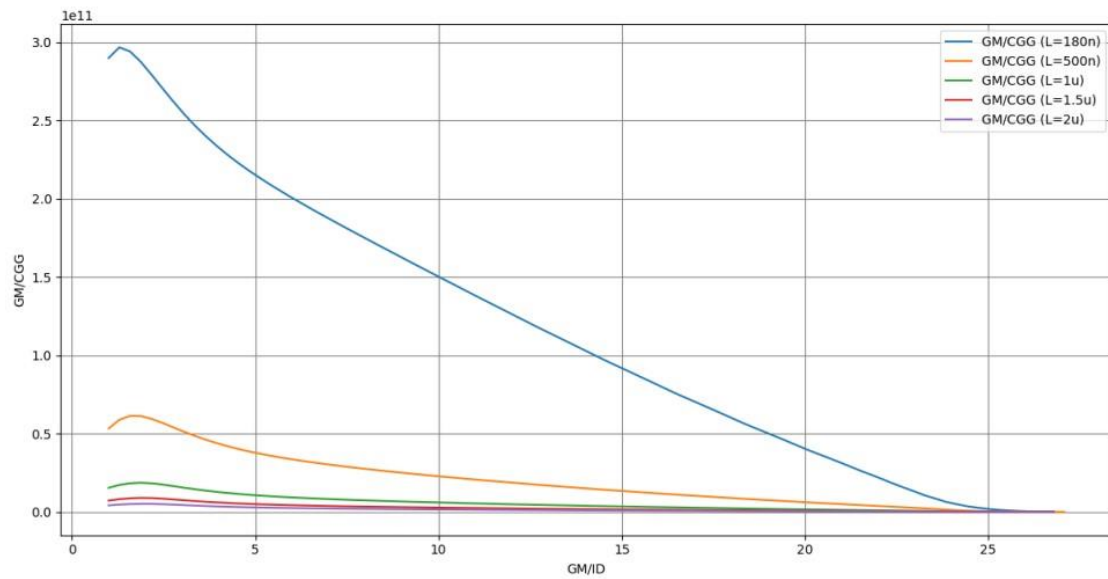
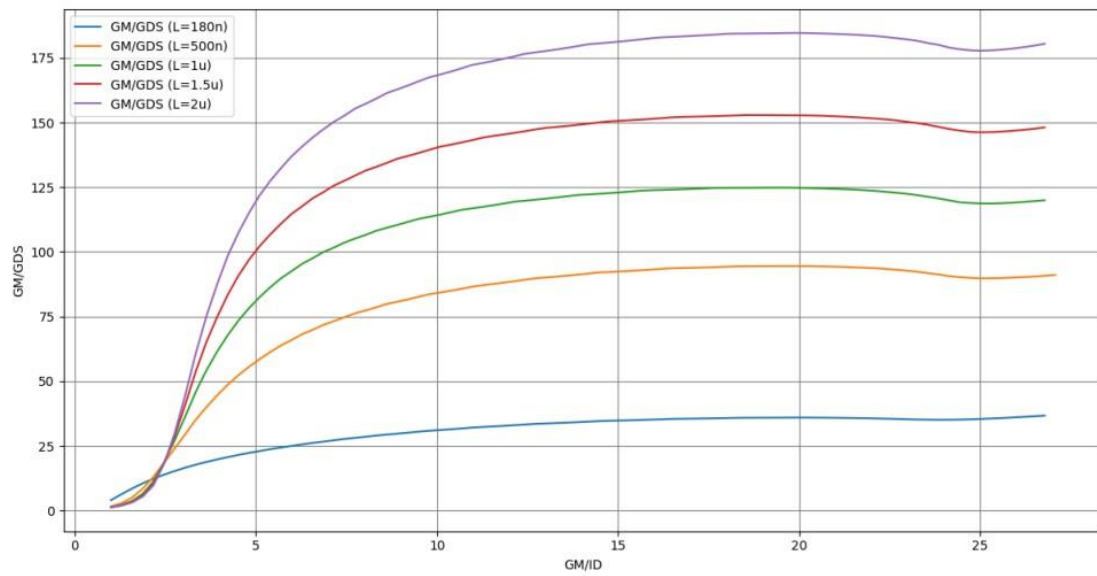
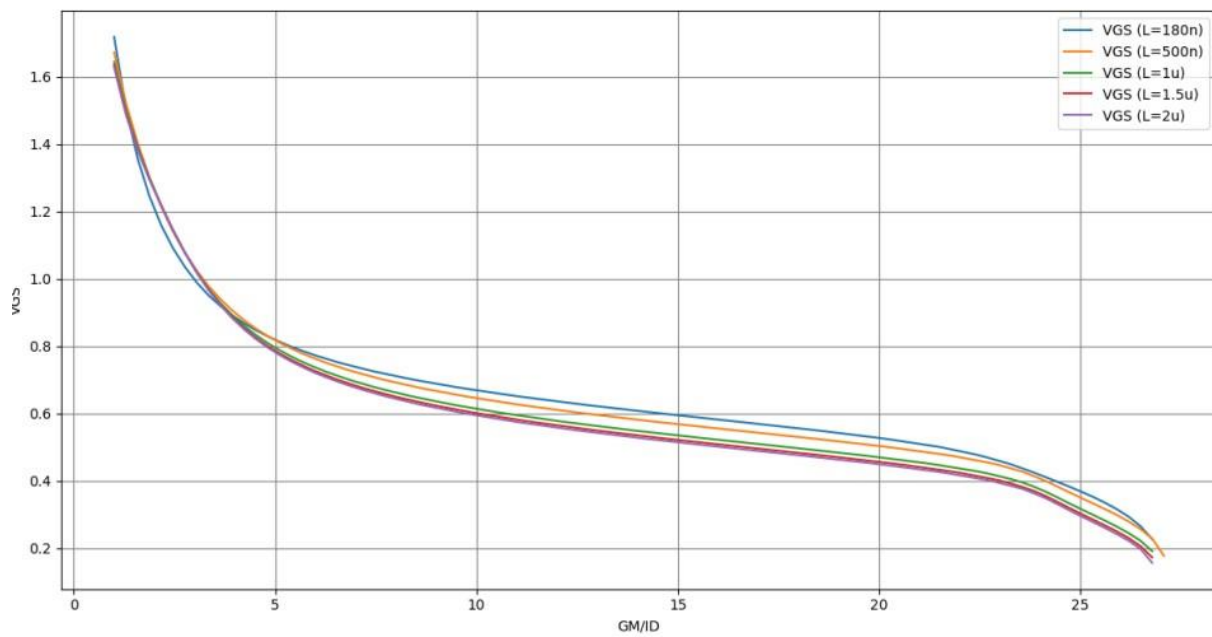
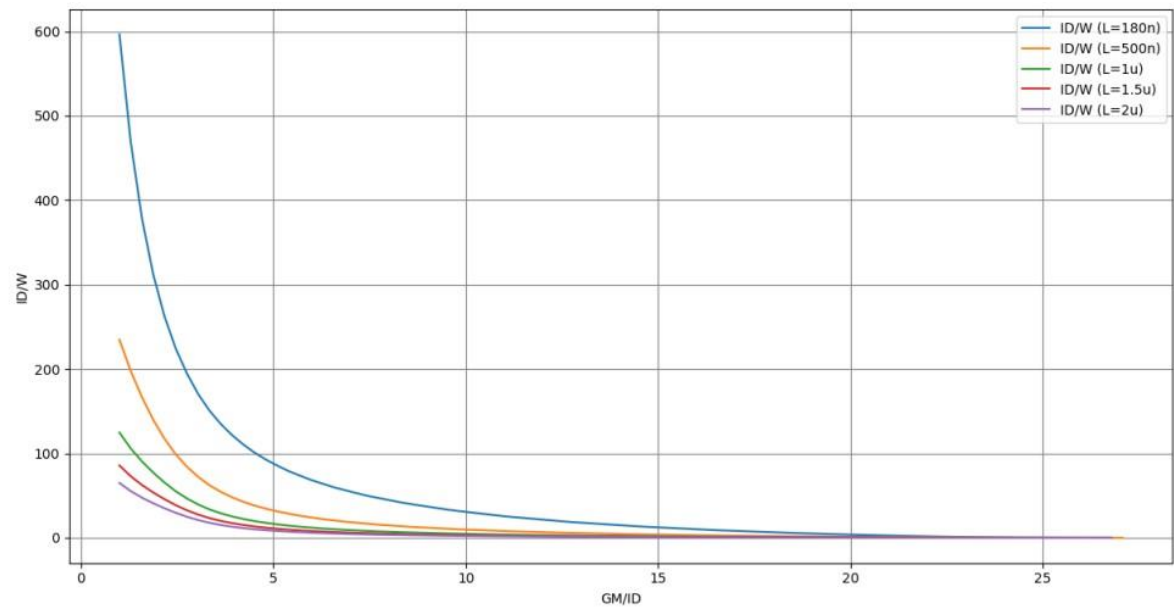


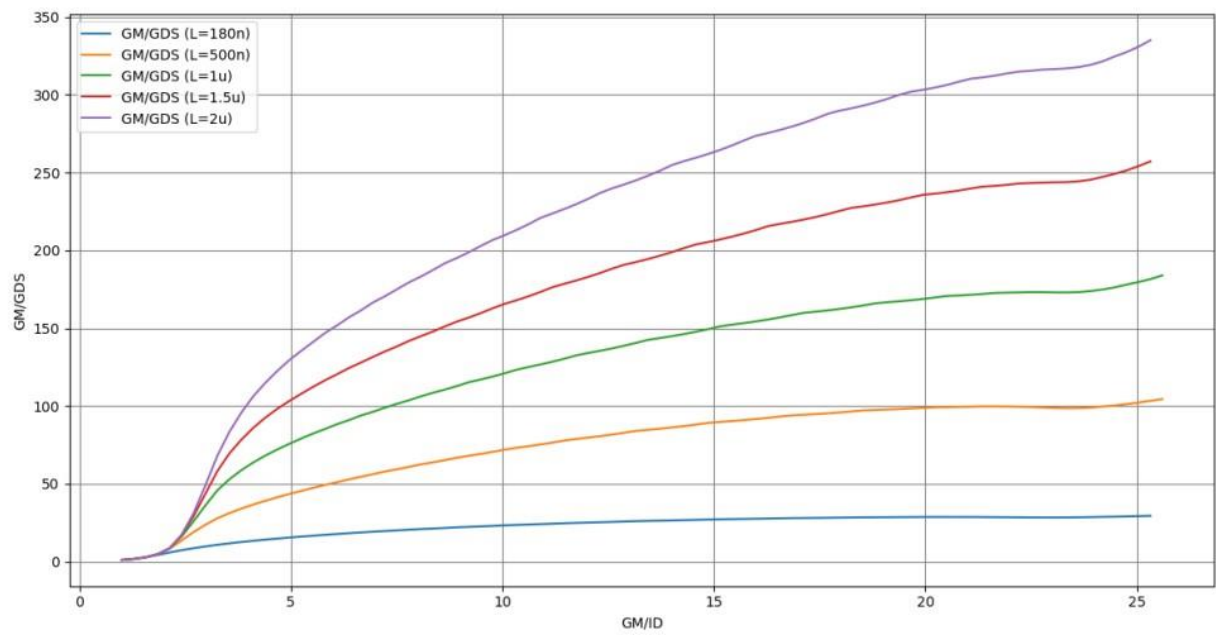
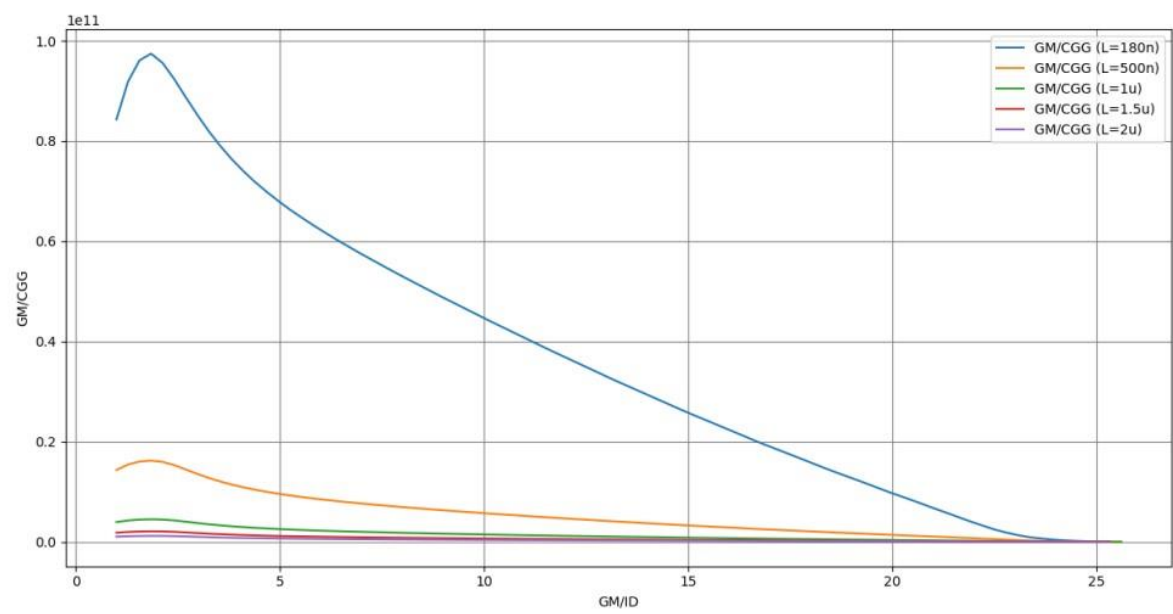
Part 1: GM/ID design charts

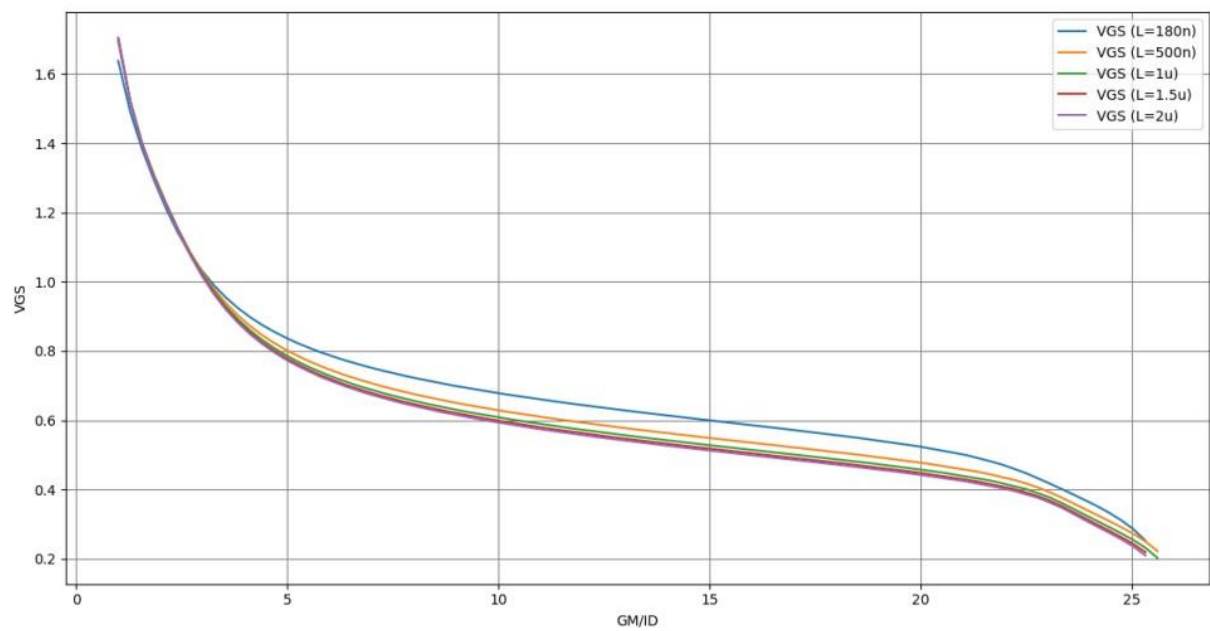
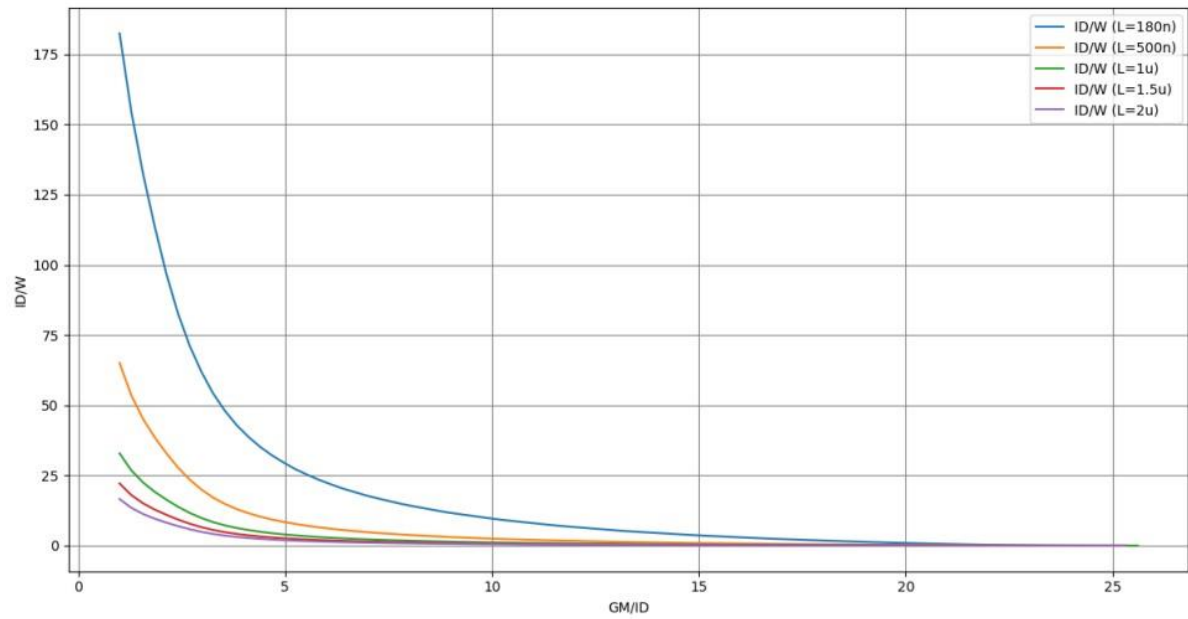
NMOS_CHARTS





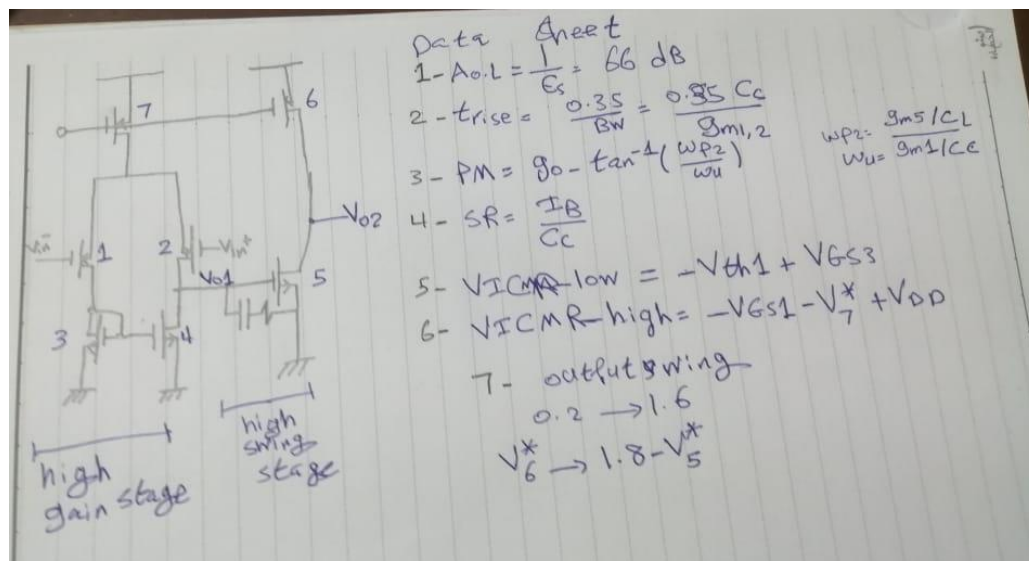
PMOS CHART





PART 2: OTA Design

DATA SHEET



Will start by assuming that first stage gain is 2 that of second stage gain

And some specs we will change to be safe like for example (current consumption will decrease it from 60 to 55 μA , CMRR will increase it to 75db and C_c will take it to be $0.5C_L$ for now)

From that we get

$$A_{v1} = g_{m1,2} * (r_{o1,2} // r_{o3,4}) = 36db$$

$$A_{v2} = g_{m5} * (r_{o6} // r_{o5}) = 30db$$

Assume $r_{o1}=r_{o2}$ we get

$$\frac{g_{m1}}{g_{ds1}} = 126$$

2- from figure 2 will take $PM=72$ (there will be a slight overshoot but will not waste as much current in stage2 if we biased them with the same g_m/ID or V_{star}) we get

Optimal Phase Margin: Summary

PM (°)	ω_{p2}/ω_{u1}	0.5% Settling	Overshoot (%)	Bandwidth	Notes
52	1	1.73x	16.3	1.27x	15.3% peaking
66	2	0.94x	4.32	1.42x	Maximally flat response No peaking Some overshoot
72	3	0.53x	0.43	1.36x	Fastest settling No peaking Very slight overshoot
76	4	0.7x	No	1.29x	Critically damped response No peaking No overshoot

$$\frac{gm2/Cl}{gm1/Cc} = 3 \rightarrow gm2 = 6gm1$$

3- we can make an educated guess to suggest that M1 and M2 we want to bias them in MI/WI for the desired high gain also we shouldn't bias them deep into weak inversion because that will decrease BW aka desired rise time might not be reached also we know that M5 in the second stage we need to consider the high swing requirement (low Vstar) to do so we will take a high gm/ID value for M5 also this will decrease wasted ID current in that stage if we connect these two guesses we can assume

$$1 - \frac{gm2}{ID2} = 1.2 \frac{gm1}{ID1} \rightarrow \text{from 2 we get } ID2 = 5ID1 = 2.5IB$$

$$2 - ID1 = \frac{55}{5+2} = 7.85\mu A \text{ (remember current consumption of } 55\mu A \text{ limitation set at the start)}$$

Slew rate total bias current condition is around $SR = \frac{IB}{Cc}$ (at $IB=15.7\mu$) if we assume $Cc=0.5 CL$ then we reach $SR=6.28$ (note that if we needed to decrease Cc further in the design procedure SR will increase so the design decision to make OTA current consumption $55\mu A$ instead 60μ is justified here)

$$3 - \text{rise time} = \frac{0.35 \cdot 2\pi \cdot Cc}{gm1} = 70ns \rightarrow gm1 = 78.5\mu S$$

$$\left(\frac{gm}{ID}\right)_1 = 10$$

4- we can now start sizing M1,M2 by assuming $V_{DS}=0.6$ and $V_{SB}=0$

NMOS1,2 SIZING1 (OTA PMOS input pair)

The screenshot shows the 'ADT Sizing Assistant' window. It has a 'Settings' button and a 'Help' button. Below them is a 'LUT Settings' section with a 'State1' dropdown and a 'Save State' button. The main area contains input fields for 'ID' (7.85u), 'gm/ID' (10), 'gm/gds' (126), 'VDS' (0.6), 'VSB' (0), and 'Stack' (1). There are 'Get' and 'Apply' buttons below these fields. A 'Y-Expr' dropdown is set to 'ID/CGG'. Below that are 'Plot', 'Replace', and 'Append' buttons. At the bottom is a 'Device Parameters' table with 28 rows.

#	Parameter	Value
1	ID	7.85u
2	L	1.06u
3	W	7.19u
4	VGS	608.5m
5	VDS	600m
6	VSB	0
7	gm/ID	9.87
8	Vstar	202.6m
9	rT	212.9M
10	gm/gds	124.6
11	VA	12.62
12	ID/W	1.092
13	gm/W	10.78
14	AREA	7.621p
15	gm	77.48u
16	gmb	24.05u
17	gds	621.8n
18	ro	1.608M
19	VTH	408m
20	VDSAT	161.1m
21	cgg	57.92f
22	cgs	50.37f
23	cgd	4.638f
24	cgb	2.921f
25	clb	7.925f
26	csb	22.44f
27	idnth2	1.132e-24
28	vgnth2	188.6e-18

From previous results we go back to DATASHEET and continue sizing NMOS3,4

From DOFS we have 1-VDS and 2-VSB and 3-rout is assumed to be same as 1,2

4-ID

5-from $CIMR-low(-v_{th1}+V_{GS3})$ we can find a value for V_{GS3} (note that this transistor will be resized for zero offset voltage meaning V_{GS3} will change later)

NMOS3,4 SIZING1 (OTA NMOS CURRENT MIRROR TRANS)

LUT
 Corner
 Temp (°C)
 State1 Save State
 ID
 VGS
 ro
 VDS
 VSB
 Stack
 Get Apply
 Y-Expr
 Plot Replace Append
 Device Parameters

#	Parameter	Value
1	ID	7.842u
2	L	1.32u
3	W	2.15u
4	VGS	608m
5	VDS	600m
6	VSB	0
7	gm/ID	9.766
8	Vstar	204.8m
9	ft	572.4M
10	gm/gds	129.9
11	VA	13.3
12	ID/W	3.647
13	gm/W	35.62
14	AREA	2.838p
15	gm	76.59u
16	gmb	20.84u
17	gds	589.5n
18	ro	1.696M
19	VTH	394.9m
20	VDSAT	159.7m

Virtuoso® ADE Assembler Editing: ic

From previous results we go back to DATASHEET and continue sizing PMOS7

From DOFS we have 1-ID 2-VDS 3-VSB

4-we can calculate the rout by using CMMRR condition

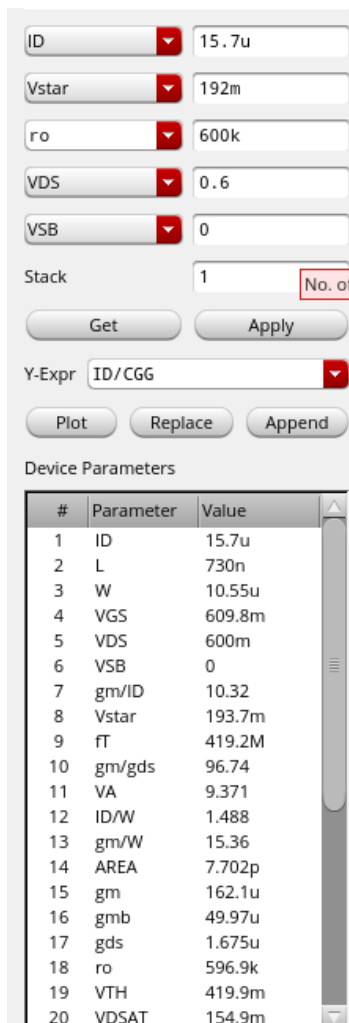
$$CMMRR = AV1 * 2gm3 * ro6 = 75db$$

$$ro6 = 600K \text{ ohms}$$

5-we can calculate Vstar with VICMR-High condition

$$Vstar7 = 1.8 - 1 - VGS1 = 192m \text{ V}$$

PMOS7 SIZING1 (HEAD CURRENT SOURCE)



Configuration window for PMOS7 SIZING1 (HEAD CURRENT SOURCE):

Parameters and values:

- ID: 15.7u
- Vstar: 192m
- ro: 600k
- VDS: 0.6
- VSB: 0
- Stack: 1
- Y-Expr: ID/CGG

Buttons: Get, Apply, Plot, Replace, Append

Device Parameters Table:

#	Parameter	Value
1	ID	15.7u
2	L	730n
3	W	10.55u
4	VGS	609.8m
5	VDS	600m
6	VSB	0
7	gm/ID	10.32
8	Vstar	193.7m
9	fT	419.2M
10	gm/gds	96.74
11	VA	9.371
12	ID/W	1.488
13	gm/W	15.36
14	AREA	7.702p
15	gm	162.1u
16	gmb	49.97u
17	gds	1.675u
18	ro	596.9k
19	VTH	419.9m
20	VDSAT	154.9m

6- we can now move to the CS stage by starting with the sizing of the top PMOS

We concluded that $ID2 = 2.5IB = 39.25\mu A$, $VDS=0.9$, $VSB=0$ and $W = 2.5W7, L=L6$

PMOS6 SIZING1 (CS CURRENT HEAD)

ID
 W
 L
 VDS
 VSB
 Stack
 Get Apply
 Y-Expr
 Plot Replace Append

Device Parameters

#	Parameter	Value
1	ID	39.25u
2	L	730n
3	W	26u
4	VGS	608.2m
5	VDS	900m
6	VSB	0
7	gm/ID	10.34
8	Vstar	193.4m
9	fT	425.7M
10	gm/gds	106.1
11	VA	10.26
12	ID/W	1.51
13	gm/W	15.61
14	AREA	18.98p
15	gm	405.9u
16	gmb	125.2u
17	gds	3.826u
18	ro	261.4k
19	VTH	419.9m
20	VDSAT	153.7m

We now move to the CS 2nd stage input MOSFET from DOFS we have that $gm/ID=12$ or that $gm_5=6gm_1$, ID, VDS and VSB

We have set the gain of the stage to be 30db from that we can calculate gds

$$30db = \frac{gm_5}{gds_5 + gds_6}$$

NMOS5(2nd stage input MOS)

ID 39.25u

gm 470u

gds 11u

VDS 0.9

VSB 0

Stack 1

Get Apply

Y-Expr ID/CGG

Plot Replace Append

Device Parameters

#	Parameter	Value
1	ID	39.25u
2	L	210n
3	W	2.09u
4	VGS	632.3m
5	VDS	900m
6	VSB	0
7	gm/ID	11.71
8	Vstar	170.8m
9	fT	15.92G
10	gm/gds	42.54
11	VA	3.633
12	ID/W	18.78
13	gm/W	219.9
14	AREA	438.9f
15	gm	459.6u
16	gmb	110.1u
17	gds	10.8u
18	ro	92.57k
19	VTH	474.5m
20	VDSAT	123.9m

The sizing

We go back to load MOSFET of the 1st stage and force a similar VGS to that of the input of the 2nd stage to avoid offset voltage .

NMOS3,4 SIZING2 (OTA NMOS CURRENT MIRROR TRANS)

State1
Save State

ID
7.85u

VGS
632m

ro
1.7M

VDS
0.6

VSB
0

Stack
1

Get
Apply

Y-Expr
ID/CGG

Plot
Replace
Append

Device Parameters

#	Parameter	Value
1	ID	7.827u
2	L	1.16u
3	W	1.56u
4	VGS	632m
5	VDS	600m
6	VSB	0
7	gm/ID	8.894
8	Vstar	224.9m
9	fT	805.8M
10	gm/gds	118.4
11	VA	13.31
12	ID/W	5.017
13	gm/W	44.62
14	AREA	1.81p
15	gm	69.61u
16	gmb	18.88u
17	gds	587.8n
18	ro	1.701M
19	VTH	399.5m
20	VDSAT	173.2m

16) Verify that your gm/ID choices do not violate the CMIR and the peak-to-peak output swing

Output swing = $V_{*5} \sim 1.8 - V_{*6} = 0.17 \sim 1.61$ ✓

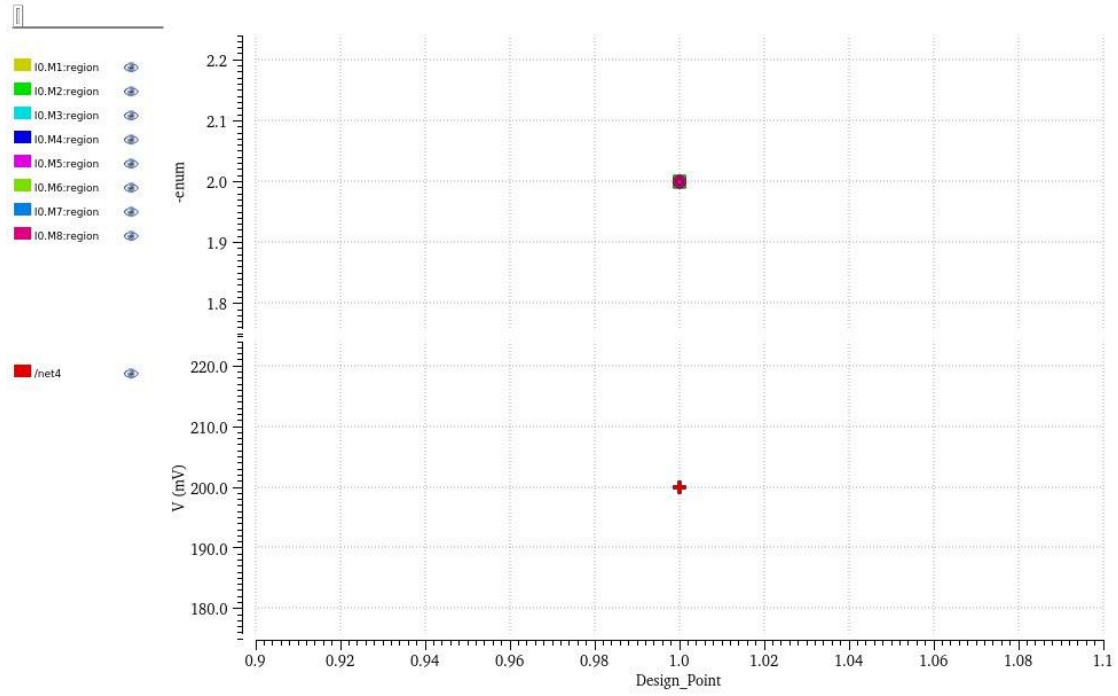
Input common mode range will verify by simulating and finding region parameter for all transistors making sure that all in saturation at

VICM=0.2

And at VICM=1

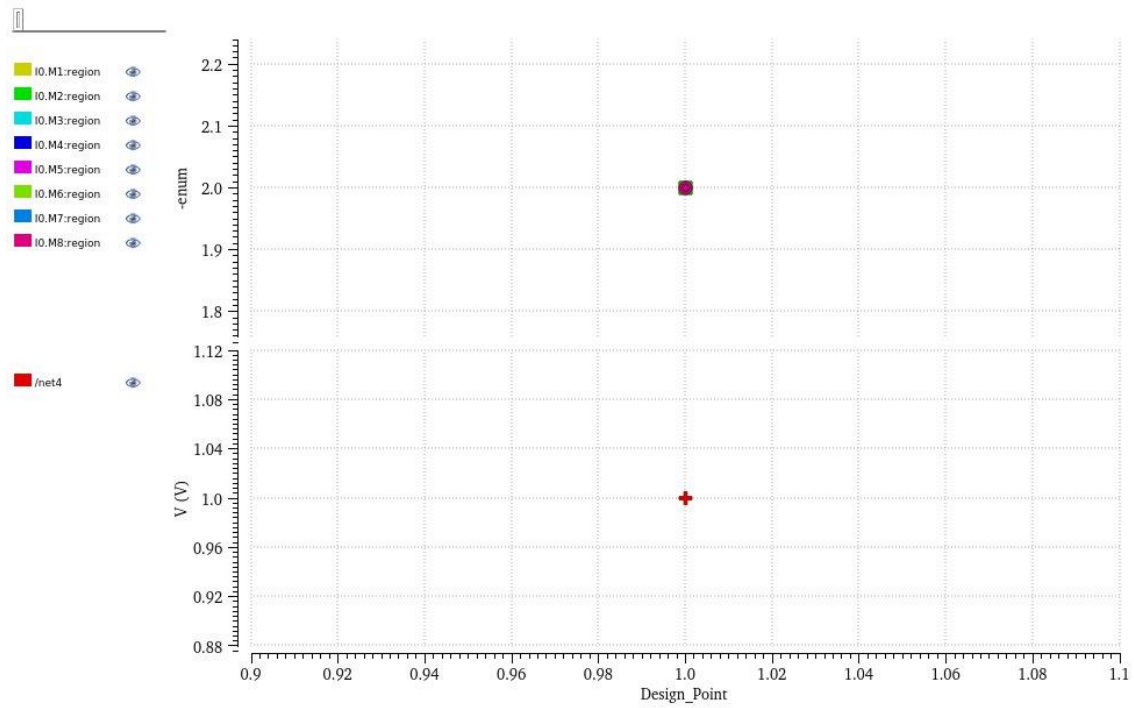
VICM=0.2 ✓

1



VICM=1 ✓

1



2) A table showing W , L , gm , ID , gm/ID , $vdsat$, $Vov = VGS - VTH$, and $V* = 2ID/gm$ of all transistors (as calculated from gm/ID curves)

Transistor	gm/ID	Length	Width	gm	ID	$Vdsat$	Vov	$V*$	function
M1, M2	10	1.06u	7.2u	77.5u	7.85u	161m	200m	202m	1 st stage input pair
M3, M4	9	1.16u	1.5u	70u	7.84u	173m	223m	225m	1 st CM load
M5	11.7	210n	2.1u	460u	39.25u	123m	157m	170m	2 nd stage input
M6	10.34	730n	26u	416u	39.25u	150m	190m	193m	2 nd stage load
M7, M8	10.4	730n	10.5u	163u	15.7u	154m	190m	193m	Bias current mirror

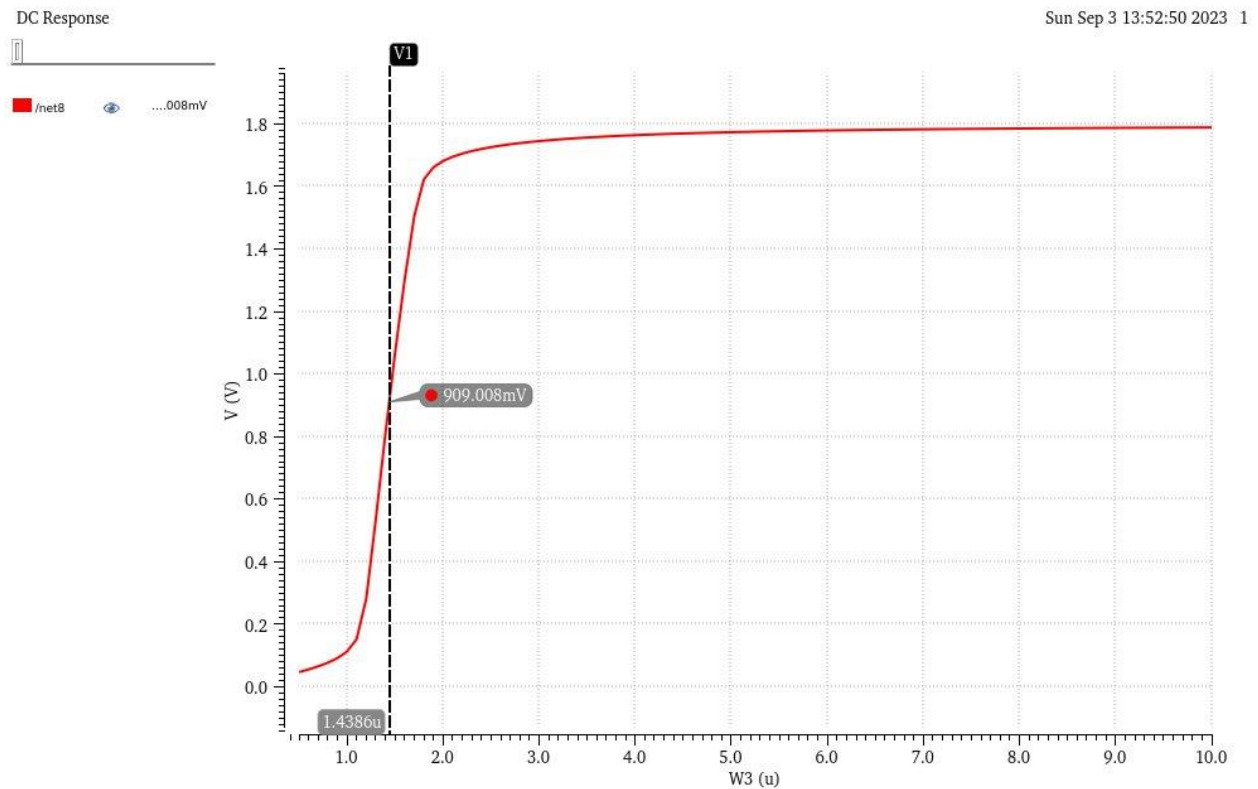
PART 3: Open-Loop OTA Simulation

Create a testbench similar to the one shown above (the shown schematic is from the 5T OTA lab). Note that IDC connection in the test bench (sinking or sourcing) may be different from the one shown above depending on the type of your input pair (PMOS/NMOS).

NOTE: The open-loop simulation will NOT work UNLESS there is ZERO offset voltage.

Report the following:

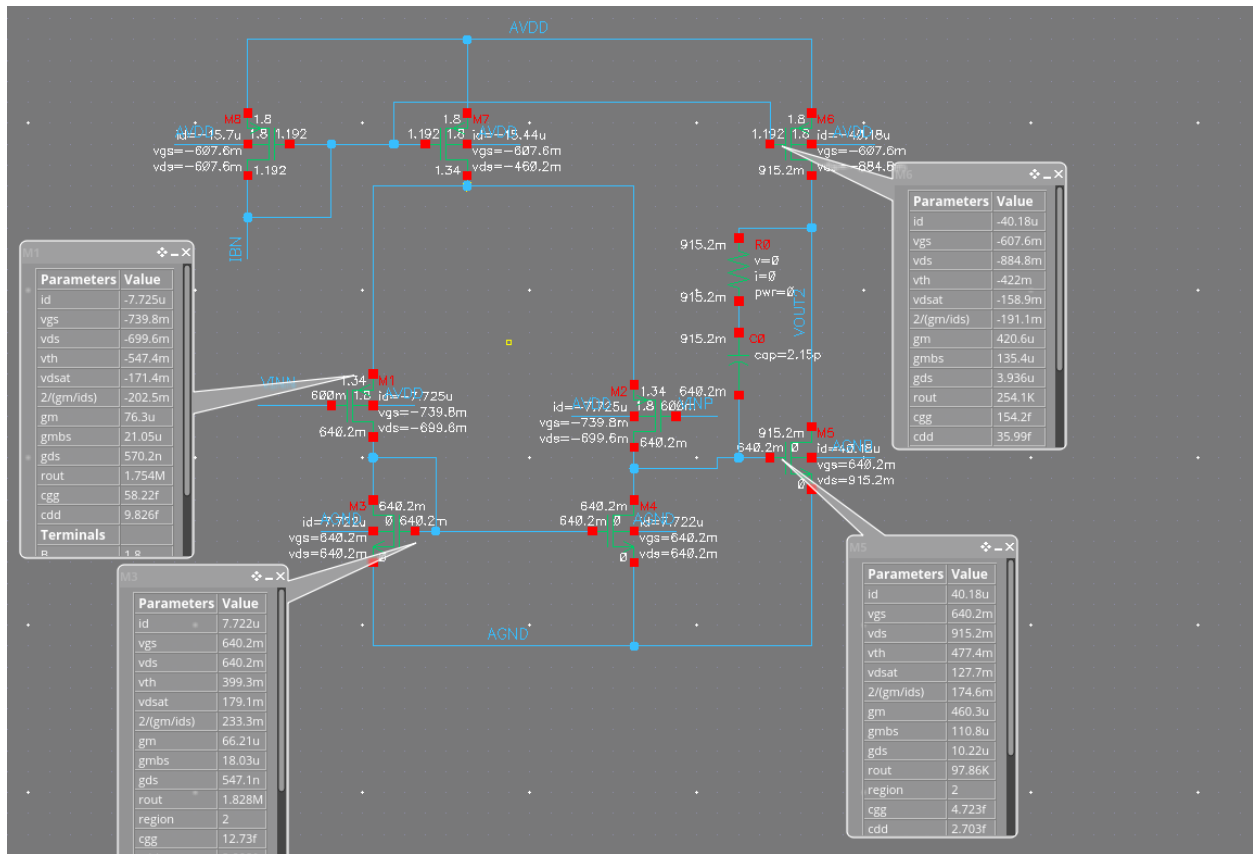
First sweeping W3 with VOUT to get $V_{OUT} = V_{DD}/2$ to ensure open loop simulation to work



We will move on with this value of W and reiterate if any requirement is violated

1) Schematic of the OTA and bias circuit with DC node voltages clearly annotated.

• Use VICM at the middle of the CMIR.



- Is the current (and gm) in the input pair exactly equal?

They are exactly equal , but we notice around 1 uA error in the second stage extra But it still doesn't violate the current consumption requirement ($40.2+15.44=55.6\text{uA} < 60\text{uA}$)

- What is DC voltage at the output of the first stage? Why?

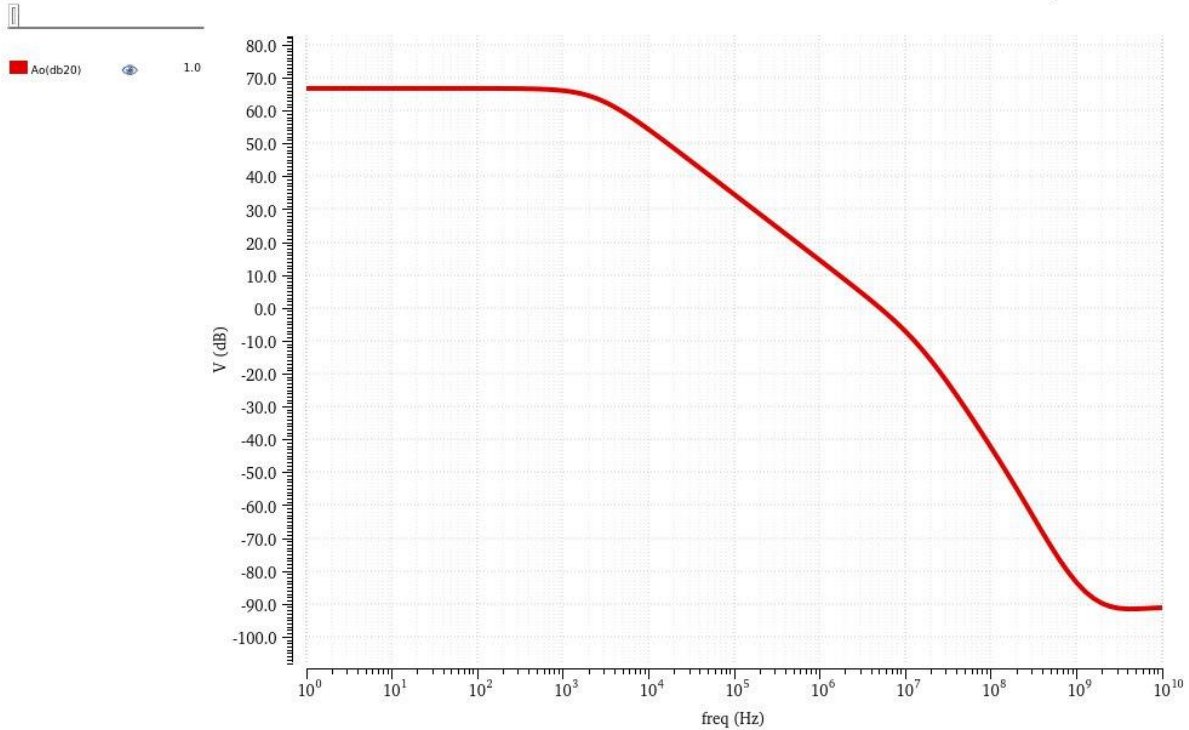
640.2mV, the DC voltage at the output of the first stage will follow the mirror node in dc simulation and that node is equal to VGS3 should be equal to VGS3 + $v_{err} \cdot A_{v1}$ (and since there is no error voltage) it is equal to VGS3 (we designed the MOSFET to have VGS of 632mV at ID=7.85u the slight error in current results in the slight change in VGS as shown)

- What is DC voltage at the output of the second stage? Why?

915mV , we biased the circuit to show such Dc voltage to maximize the swing

1) Diff small signal ccs:

- Use AC analysis (1Hz:10Gz, logarithmic, 10 points/decade).
- Set VIDAC = 1 and VICMAC = 0.
- Use VICM at the middle of the CMIR.
- Plot diff gain (in dB) vs frequency.



Test	Output	Nominal
Filter	Filter	Filter
icDesign02_lab_...	Ao(db20)	
icDesign02_lab_...	Aomax(db20)	66.89
icDesign02_lab_...	BW1	2.434K
icDesign02_lab_...	GBW1	5.394M
icDesign02_lab_...	UGF1	5.113M
icDesign02_lab_...	VOUT	

- Compare simulation results with hand calculations in a table

Hand analysis:

$gm1=76.5\mu S$, $ro1=1.75M\ ohms$, $gm5=470\mu S$ $ro5=100k\ ohms$, $ro6=250k\ ohms$,
 $C_{gg}=60fF$

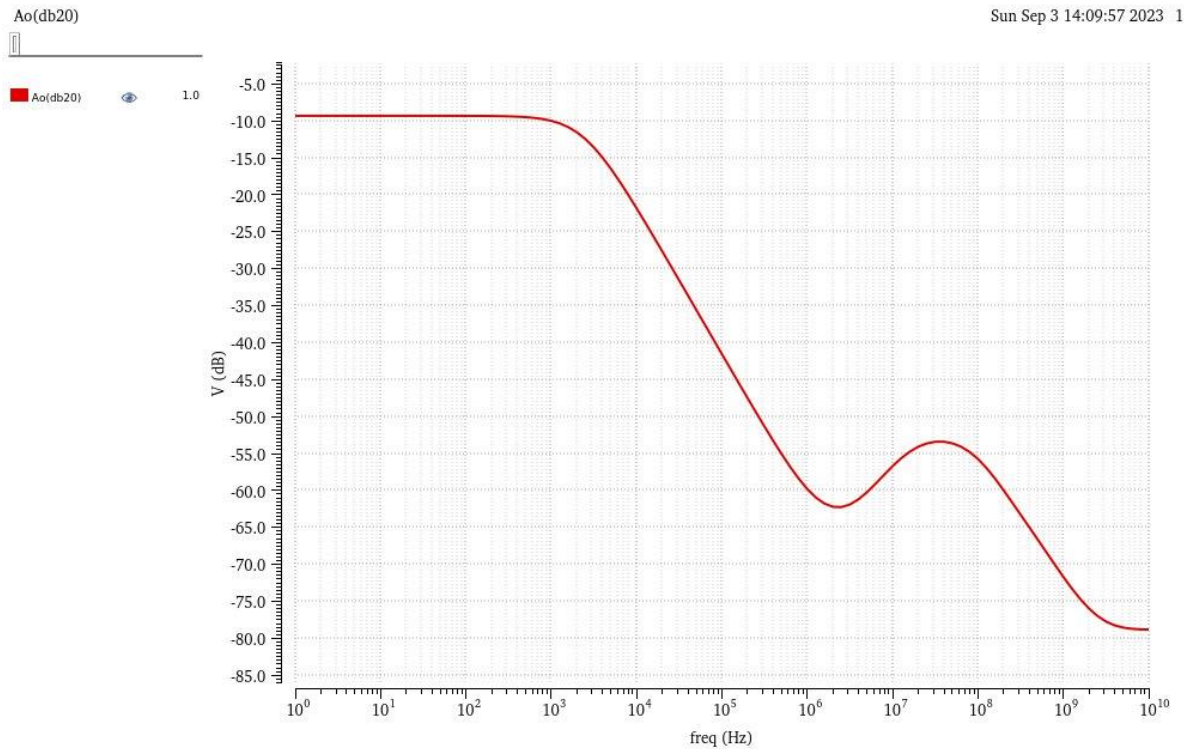
$$DCgain = \frac{gm1ro1}{2} * gm3(ro3//ro6) = 2.14k = 66.8db$$

$$GBW = \frac{gm1}{2\pi * 0.43(CL + 2cdd)} = 5.5Mhz$$

	CAD	Hand analysis
DCgain	66.9 db	66.8 db
GBW	5.4M Hz	5.5M Hz

2) CM small signal ccs:

- Use AC analysis (1Hz:10Gz, logarithmic, 10 points/decade).
- Set VICMAC = 1 and VIDAC = 0.
- Use VICM at the middle of the CMIR.
- Plot CM gain in dB vs frequency.



- Compare simulation results with hand calculations in a table.

Hand analysis:

$gm_1=76.5\mu S$, $r_{out1}=1.75M\ \Omega$, $gm_5=460\mu S$, $r_{out5}=100k\ \Omega$, $r_{out6}=250k\ \Omega$,
 $C_{gg}=60fF$, $gm_5=66\mu S$, $r_{out5}=550k\ \Omega$

$$AV_{cm} = \frac{1}{2gm_3R_{ss}} * gm_5(ro_6//ro_5) = 0.7 = -7db$$

	CAD	Hand analysis
AV_{cm}	-8 db	-7 db

3) (Optional) CMRR:

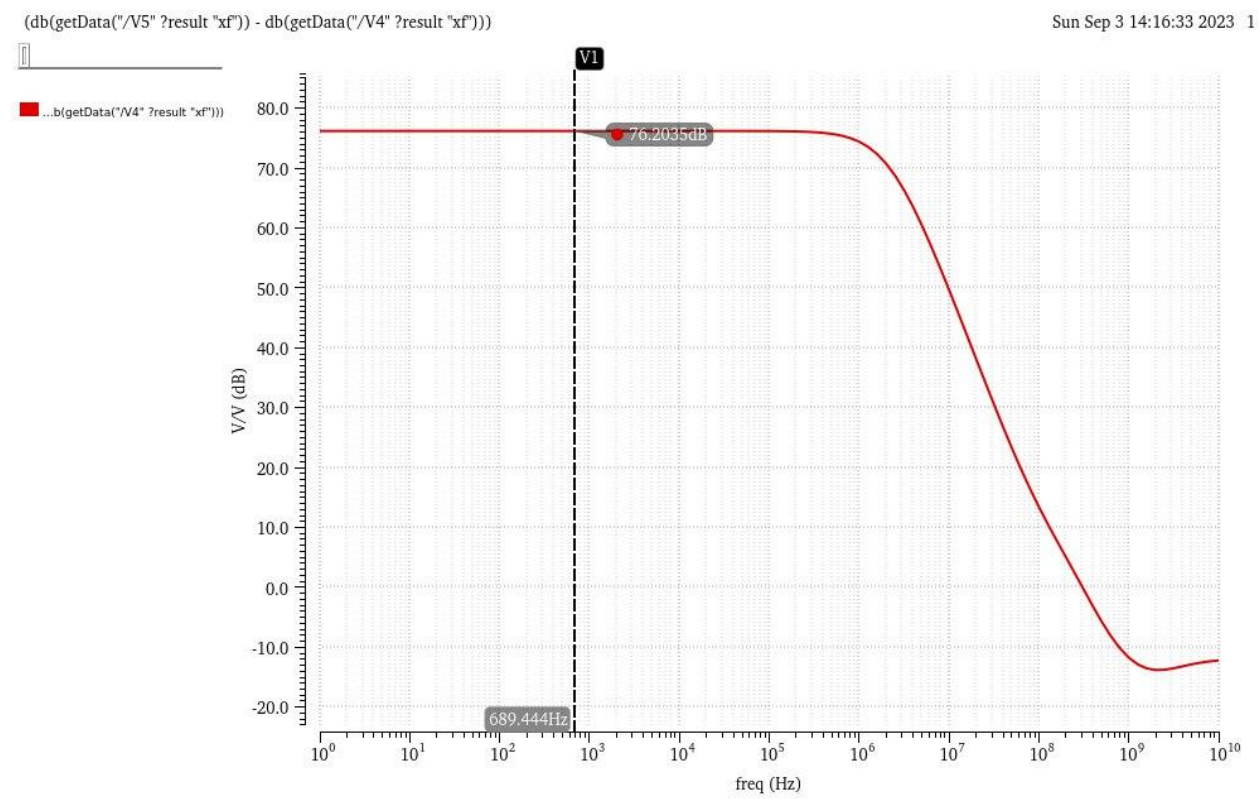
→ Cadence Hint: In Mentor Pyxis you have to get Avd and Avcm from two independent simulation runs (we cannot run both simultaneously because we have single ended output, thus we cannot differentiate between diff and CM signals at the output). But for Cadence Virtuoso you should use XF analysis

(1Hz:10Gz, logarithmic, 10 points/decade) because you need to calculate the transfer function between multiple inputs and a single output.

→ Cadence Hint: Access XF analysis results from the results browser or from adexl results tab (Right Click -> Direct Plot -> Main Form). You may use this expression in the calculator to plot the CMRR: $\text{dB20}(\text{mag}(\text{getData}("/V2" ?result "xf"))) - \text{dB20}(\text{mag}(\text{getData}("/V1" ?result "xf"))) .$

• Use VICM at the middle of the CMIR.

• Plot CMRR in dB vs frequency at VICM at the middle of the CMIR.



• Compare simulation results with hand calculations in a table.

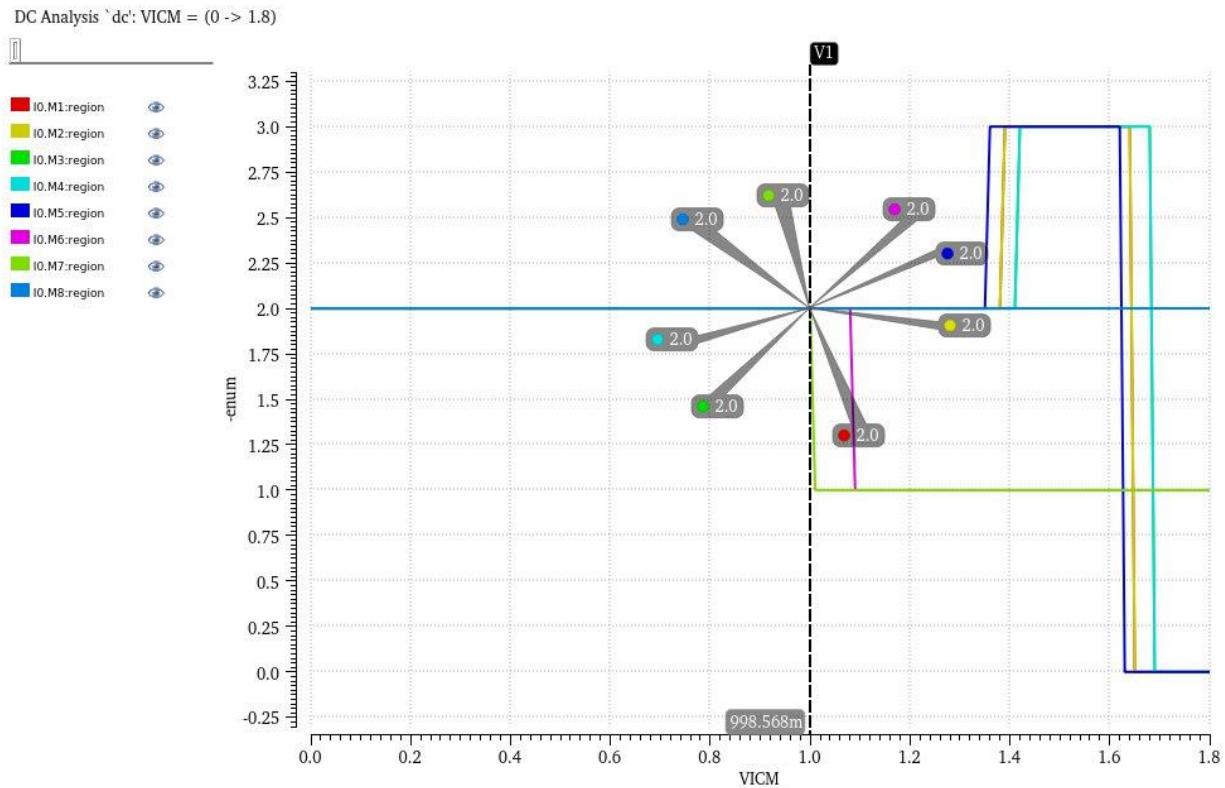
Hand analysis

$$\text{CMRR} = \frac{gm1 * ro1}{2} * 2gm3 * ro7 = 75\text{db}$$

	CAD	Hand analysis
CMRR	76	75

4) CM large signal ccs (region vs VICM):

- Use DC sweep (not parametric sweep) VICM = 0:10m:VDD.
- Plot “region” OP parameter vs VICM for the input pair and the tail current source (0 cut-off, 1 triode, 2 sat, 3 subth, and 4 breakdown).
- Plot “region” OP parameter vs VICM for the input pair and the tail current source.



- Find the CM input range (CMIR). Compare with hand analysis in a table.

CMIR-low=0

CMIR-high=1

Hand analysis:

$$VICM - low = -vth1 + VGS3 = -0.54 + 0.64 = 0.1$$

$$VICM - high = 1.8 - Vstar7 - VGS1 = 1.008$$

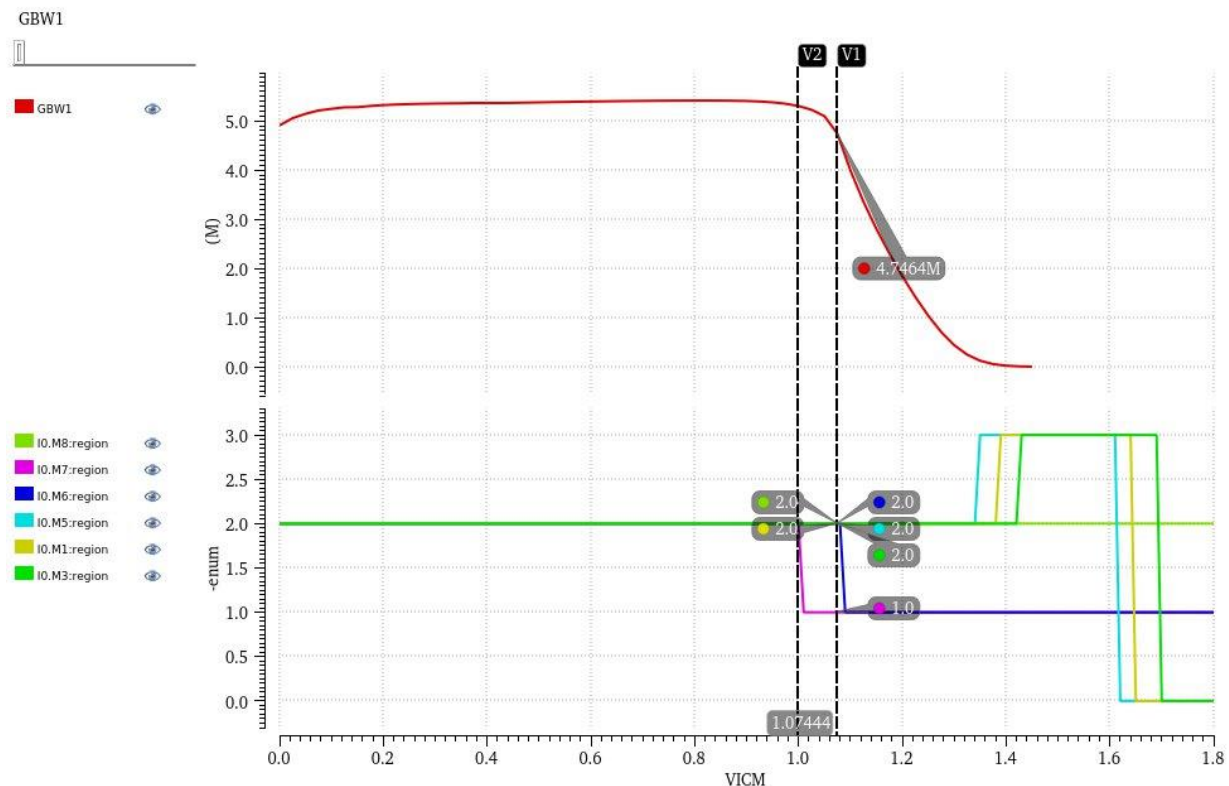
	CAD region	Hand analysis
VICM-low	0	0.1
VICM-high	1	1.008

- Note that the drawback of this method is that the “region” parameter cannot be experimentally

measured in the lab.

5) (Optional) CM large signal ccs (GBW vs VICM):

- Use AC analysis (1Hz:10Gz, logarithmic, 10 points/decade).
- Set VIDAC = 1 and VICMAC = 0.
- Use parametric sweep (not DC sweep) VICM = 0:25m:VDD.
- Plot GBW vs VICM. Plot the results overlaid on the results of the previous method (region parameter).



- Annotate the CM input range. Calculate the input range as the range over which the GBW is within 90% of the max GBW, i.e., 10% reduction in GBW3.

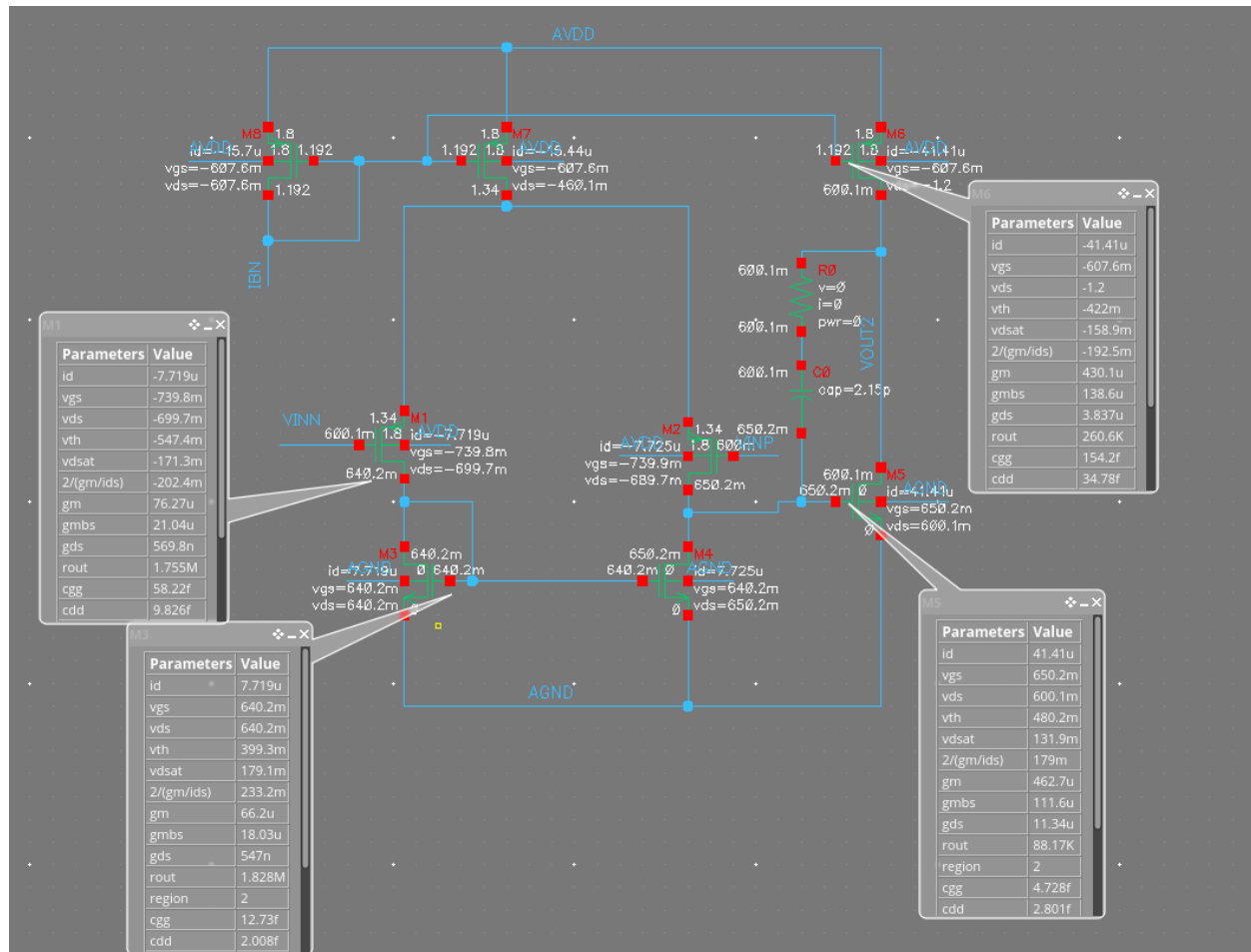
PART 4: Closed-Loop OTA Simulation

Create a new testbench with the OTA connected in a unity gain buffer feedback configuration (the shown schematic is from the 5T OTA lab). Place a current probe (iprobe) or a zero voltage source in the feedback loop.

Report the following:

1) Schematic of the OTA and the bias circuit with DC OP point clearly annotated in unity gain buffer configuration.

• Use VICM at the middle of the CMIR.



• Are the DC voltages at the input terminals of the op-amp exactly equal? Why?

There is a 0.1 mV difference between the two, there will always be a small Verr and that is equal to $\Delta v_{out1} = v_{err} * A_{v1}$

• Is the DC voltage at the output of the first stage exactly equal to the value in the open-loop simulation? Why?

Very close to it but it is not , open loop simulation there was no Verr between the two inputs so the increase in the voltage at the output of the first stage is due to that Verr that will be multiplied by first stage gain

• Is the current (and gm) in the input pair exactly equal? Why?

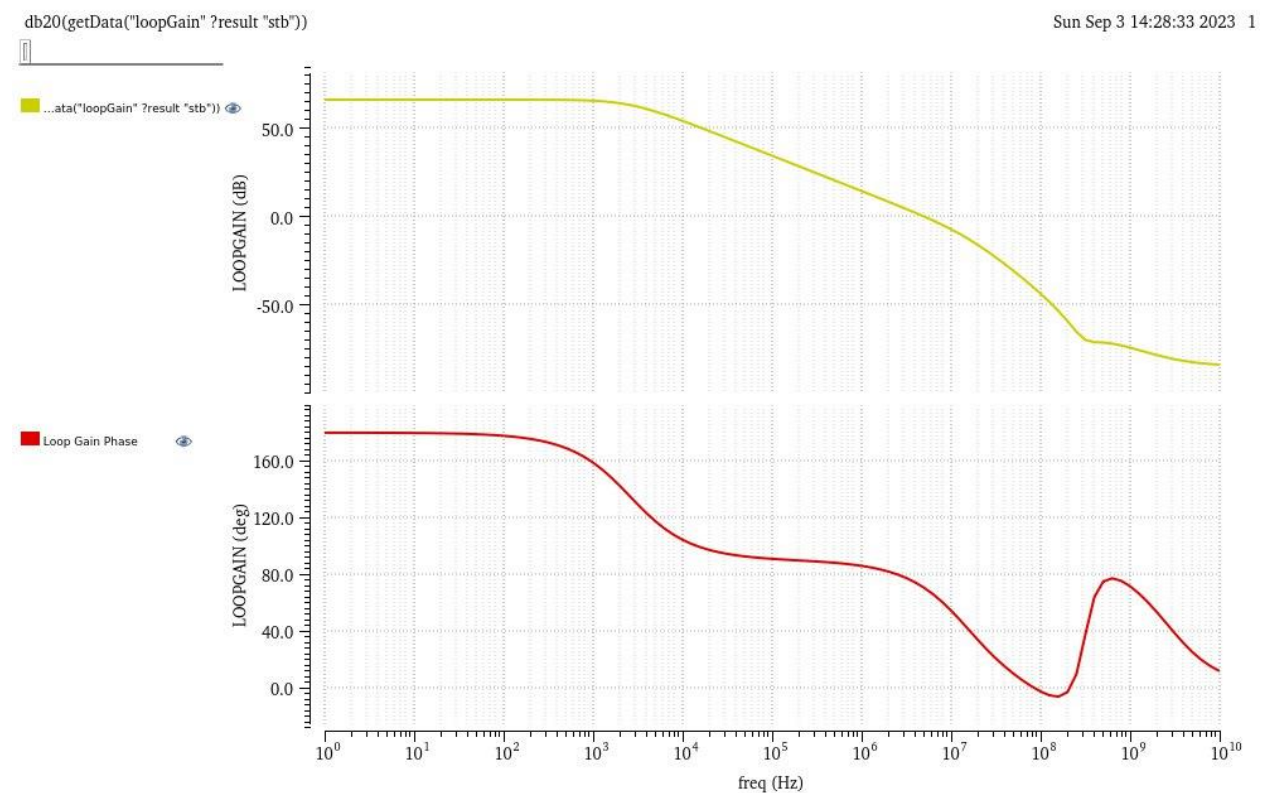
Very very slight difference , mismatch between the two branches

2) Loop gain:

• Use STB analysis (1Hz:10Gz, logarithmic, 10 points/decade) in unity gain buffer configuration.

• Use VICM at the middle of the CMIR.

• Plot loop gain in dB and phase vs frequency.



→ Cadence Hint: Access STB analysis results from the results browser or from adexl results tab (Right Click in adexl Results tab-> Direct Plot -> Main Form).

Test	Output	Nominal	
Filter	Filter	Filter	Filter
icDesign02_lab_...	db20(getData("l...		
icDesign02_lab_...	Loop Gain Phase		
icDesign02_lab_...	Loop Gain Magn...		
icDesign02_lab_...	Ao_maxdb20	66.34	
icDesign02_lab_...	Phase Margin	70.37	
icDesign02_lab_...	BW	2.59K	
icDesign02_lab_...	GBW	5.388M	
icDesign02_lab_...	UGF	5.1M	

- Compare DC gain, fu, and GBW with those obtained from open-loop simulation. Comment

	OL CAD	CL CAD
DC gain	66.9 db	66.34 db
GBW	5.39M Hz	5.39M Hz
UGF	5.1M Hz	5.1M Hz

No difference between two simulations if OL biased correctly

- Report PM. Compare with hand calculations. Comment.

Hand calculations for phase margin ($C_c=0.43C_L$, $gm_1=76.5\mu S$, $gm_2=460\mu S$, $C_L=5pF$)

$$PM = 90 - \tan^{-1}\left(\frac{w_u}{w_{p2}}\right)$$

- Compare simulation results with hand calculations in a table.

	CAD	Hand analysis
PM	70.3	69

3) Slew rate:

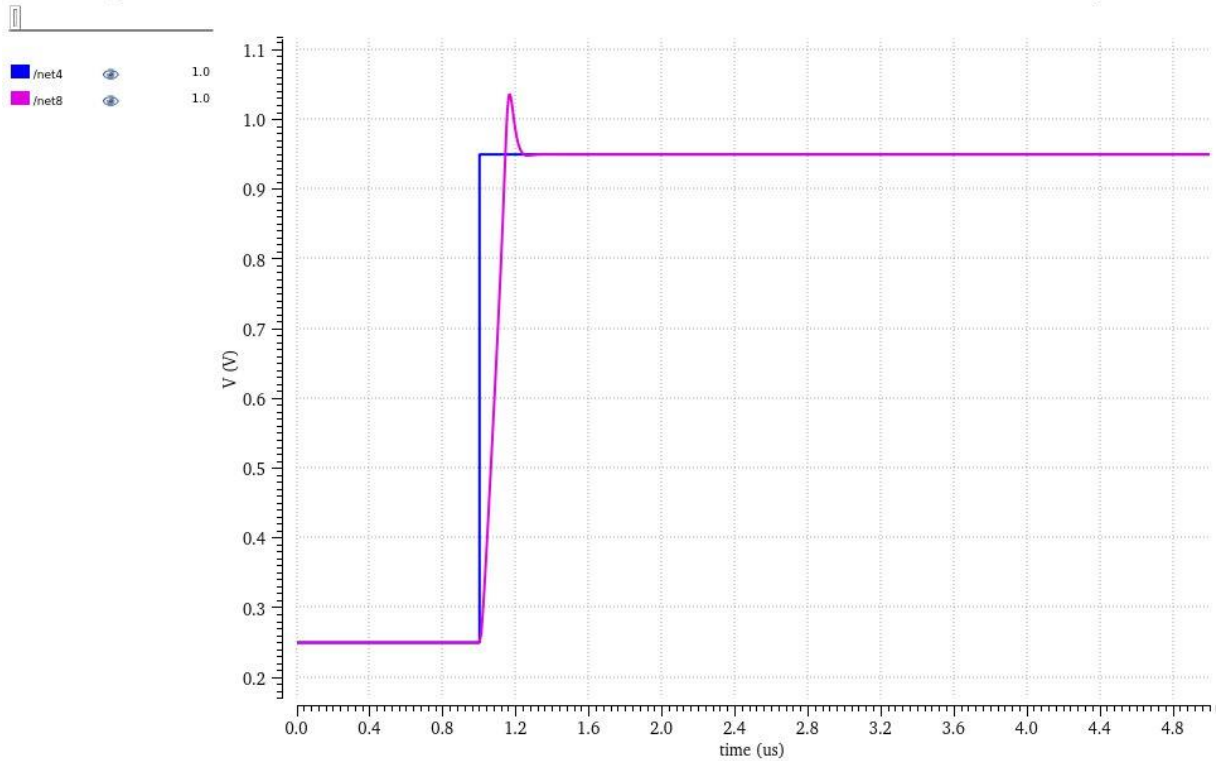
- Apply a step input with the following parameters (delay = 1us, initial value = CMIR-low + 50mV, final

value = CMIR-high – 50mV, rise time = 1ns, period = 1s, width = 1s). Note that we want a single step

input, which is why we selected very large period and width for the pulse.

- Run transient analysis (stop = 5us and step = 0.1ns).

- Report Vin and Vout overlaid.



- Report the slew rate.



SR=5.25 V/us

- Compare simulation results with hand calculations in a table.

Hand calculations slew rate

From simulation results $I_B = 15.2 \mu A$, $C_c = 0.43 \cdot C_L$

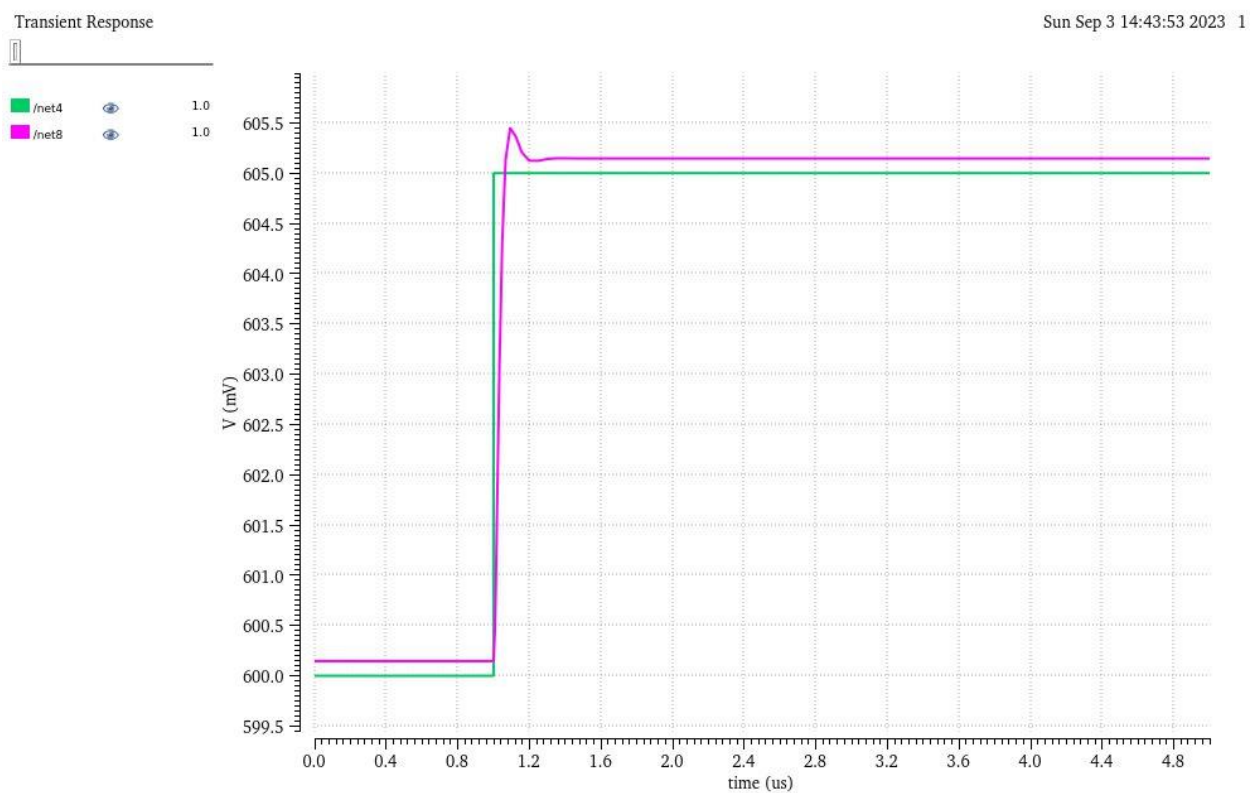
$$SR = \frac{I_B}{C_c} = \frac{15.2}{2.15} = 7 \mu S$$

	CAD	Hand analysis
SR	5.25	7

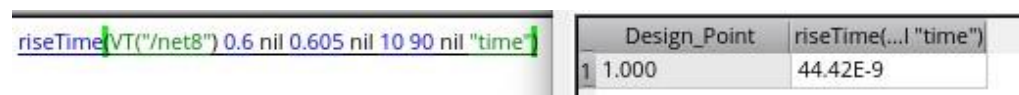
4) Settling time:

• Apply a small signal step input with the following parameters (delay = 1us, initial value = the middle

of the CMIR, final value = the middle of the CMIR + 5mV, rise time = 1ns, period = 1s, width = 1s). Note that we want a single step input, which is why we selected very large period and width for the pulse. Note that we apply a small signal pulse (5mV step) to measure the small signal settling time.



• Calculate the output rise time from simulation.



• Compare simulation results with hand calculations in a table4

From simulation results $g_m = 76.5 \mu S$, $C_c = 2.15 pF$

$$rise\ time = \frac{0.35 * 2\pi * 2.15 \times 10^{-12}}{76.5 \times 10^{-6}} = 60 ns$$

	CAD	Hand analysis
Rise time	45n S	60n S

• Do you see any ringing? Why?

There is no ringing although there is a very slight overshoot, from simulation results PM is around 70 degrees and this PM shouldn't result in any relevant ringing in time domain or peaking in frequency domain. (note this we can improve this phase margin to 72 even 73 degrees and eliminate this overshoot even further by increasing the value of C_c since we have a small leeway in the GBW and rise time results)